

POTHOLELOG · REQUIREMENTS

PotholeLog requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

PotholeLog requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish, and will be checked by calculation at TRL 3 and by field data later. The status column gives the concept's position today: "met by design" means the chosen parts and architecture meet the target on paper; "unverified" means only data from a vehicle can show it.

Table 1. Requirements.

ID	Requirement	Target	Status at TRL 2	Verification (TRL 3 or later)
R1	Detect potholes and severe defects in the wheel path	80 % or more of defects at least 50 mm deep and 300 mm long found within 3 passes	Unverified	Detection model on logged data against a surveyed defect list
R2	Keep false reports low	10 % or fewer of reported defect clusters (2 or more passes) are not real defects	Unverified; Pothole Patrol reported over 90 % real after clustering	Field check of reported clusters
R3	Report roughness per road segment	One roughness value per 100 m segment per pass, calibrated per vehicle to an IRI estimate; r^2 of 0.8 or better against reference IRI on calibration sections	Unverified; needs reference sections from a partner	Calibration runs following World Bank Technical Paper 46
R4	Locate defects	Defect cluster within 10 m of its true position (95 %) and on the correct street	At risk in dense urban canyons	GNSS logs against surveyed points
R5	Work across normal fleet speeds	Roughness valid from 20 to 80 km/h; defect detection from 10 to 80 km/h	Partly met; segments driven below 20 km/h give no roughness value	Speed-banded calibration
R6	Sample fast enough	3-axis acceleration at 400 Hz or more, range ± 8 g or more; GNSS position at 5 Hz or more	Met by design (400 Hz, ± 16 g, 10 Hz GNSS)	Datasheets and firmware configuration
R7	Store data on the vehicle	30 days or more of raw data and summaries without upload	Met by design (about 5 months on 32 GB, estimate)	Storage calculation
R8	Get data off the vehicle	Summaries uploaded automatically within 24 h, with no driver or staff action	Met by design (depot Wi-Fi); real-time upload not provided	Design review
R9	Run from vehicle power	9 to 36 V DC input (12 V and 24 V systems); reverse-polarity protected; transient levels of ISO 16750-2 as a target; under 1 W running; no draw with ignition off	Met by design except transient levels, which are unverified	Power budget; later bench test
R10	Shut down cleanly	Files closed without corruption when power is cut at any time	Met by design (about 9 s hold-up, estimate)	Hold-up calculation
R11	Survive the vehicle environment	Operating -20 to +70 °C; enclosure IP65; no loosening under vehicle vibration	Enclosure met by design; temperature and vibration unverified	Component ratings; later vibration test
R12	Install quickly and safely	30 min or less by a fleet mechanic; no cutting or welding of structural members; bracket first natural frequency above 150 Hz	Install time and bracket stiffness unverified	Design review; TRL 3 calculation of plate stiffness
R13	Protect privacy	No camera or microphone; published data limited to road-segment roughness and defect locations, with no vehicle	Met by design	Design review; data schema review

ID	Requirement	Target	Status at TRL 2	Verification (TRL 3 or later)
		tracks, timestamps or driver identity; no driver scoring		
R14	Open outputs	CSV and GeoJSON outputs that CityTwin and common GIS tools can read	Met by design (format proposed, awaiting Amish)	Sample export
R15	Low cost and buildable	Parts \$70 or less per unit; off-the-shelf modules; no custom PCB	Met, with \$1 margin (about \$69, indicative)	Priced BOM
R16	Fit the fleet types in the pitch	Buses and refuse trucks; bicycles	Not met for bicycles: the concept needs vehicle power and a rigid floor; a bicycle needs its own battery and mount	Design review

■ Requirements not met or at risk

- **R16 (bicycles) not met.** The concept runs from vehicle power. A bicycle variant needs a battery, a charger and a rider-independent mount, and cyclist body motion would need its own calibration. Proposed, awaiting Amish: treat bicycles as a later variant.
- **R3 and R1 unverified.** Both depend on field data and a calibration partner with reference IRI sections.
- **R4 at risk.** Single-frequency GNSS in streets lined with tall buildings can be several meters to tens of meters off. Clustering over many passes helps, but this is unproven.
- **R5 partly met.** Stop-and-go traffic leaves some segments without a roughness value on a given pass.
- **R15 margin is thin.** Any cellular option (about \$20 to \$30 more for a module, plus a data plan) would exceed the \$70 budget.

■ Assumptions

- Host vehicles run about 200 km a day (estimate for an urban bus or refuse truck) and return to a depot with Wi-Fi each night.
- The logger is mounted on the sprung mass (floor or body structure), not the axle; the suspension filters the input, which calibration corrects.
- 100 m segments match common practice for network roughness reporting; defects are reported as point events.
- A partner agency can provide reference IRI on a few calibration sections, or allow a rod-and-level survey of them.